ML Visualisation Tool

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GitHub repo: https://github.com/MuffinMufido/HealthML.wiki

Live demo URL: http://127.0.0.1:5173

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Sprint Focus: Full pipeline completion (Steps 6 & 7), explainability quality, ethics/bias governance, and Sprint 4 submission compliance.

1) Burndown Chart

Sprint 4 burndown is attached from Jira to show remaining story points against the ideal line. The chart confirms that Step 6/7 implementation and integration tasks were completed by the sprint close window.

- Attach Jira burndown screenshot in final submission packet.

2) Completed This Week

1. Step 6 Explainability finalized: feature importance chart, clinical labels, patient-level explanation, amber caution and blue what-if banner.
2. Step 7 Ethics & Bias finalized: subgroup table, bias status tags, checklist with progress bar, training representation warning, and case-study cards.
3. Certificate backend integration completed and validated using active branch endpoints.
4. End-to-end flow (Step 1 through Step 7) revalidated using the heart_failure_clinical_records_dataset.csv dataset.

3) In Progress

- No Sprint 4 core functional blocker remains in accepted scope.
- Optional cleanup in progress: automation script alignment with latest UI text/selector changes.
- Optional environment hardening: Docker daemon availability check for reproducible compose run.

4) Blocked / At Risk

- Risk: Legacy Playwright assertions can fail due outdated expectations, even when product behavior is correct.
- Mitigation: Treat as test-maintenance item; update selectors and expected strings in next maintenance pass.
- Risk: Docker runtime check depends on local daemon state.
- Mitigation: Run final compose verification on a machine with active Docker Desktop daemon.

5) Key Decisions Made

1. Accepted current ai_implementation branch as Sprint 4 baseline after live service and API revalidation.
2. Classified mismatched automation assertions as non-blocking test drift rather than product defects.
3. Prioritized submission readiness: functional acceptance + evidence completeness + report traceability.

6) Test Results

- Frontend health: PASS (http://127.0.0.1:5173)
- Node backend health: PASS (/health on port 3001)
- Python ML service health: PASS (/health on port 8000)
- E2E API flow (upload->map->prepare->train->explain->fairness->certificate): PASS
- Step 6 acceptance behavior: PASS
- Step 7 acceptance behavior: PASS
- Certificate endpoint behavior in current branch: PASS

Evidence Screenshots (Live System)

Step 1 — Clinical Context selection

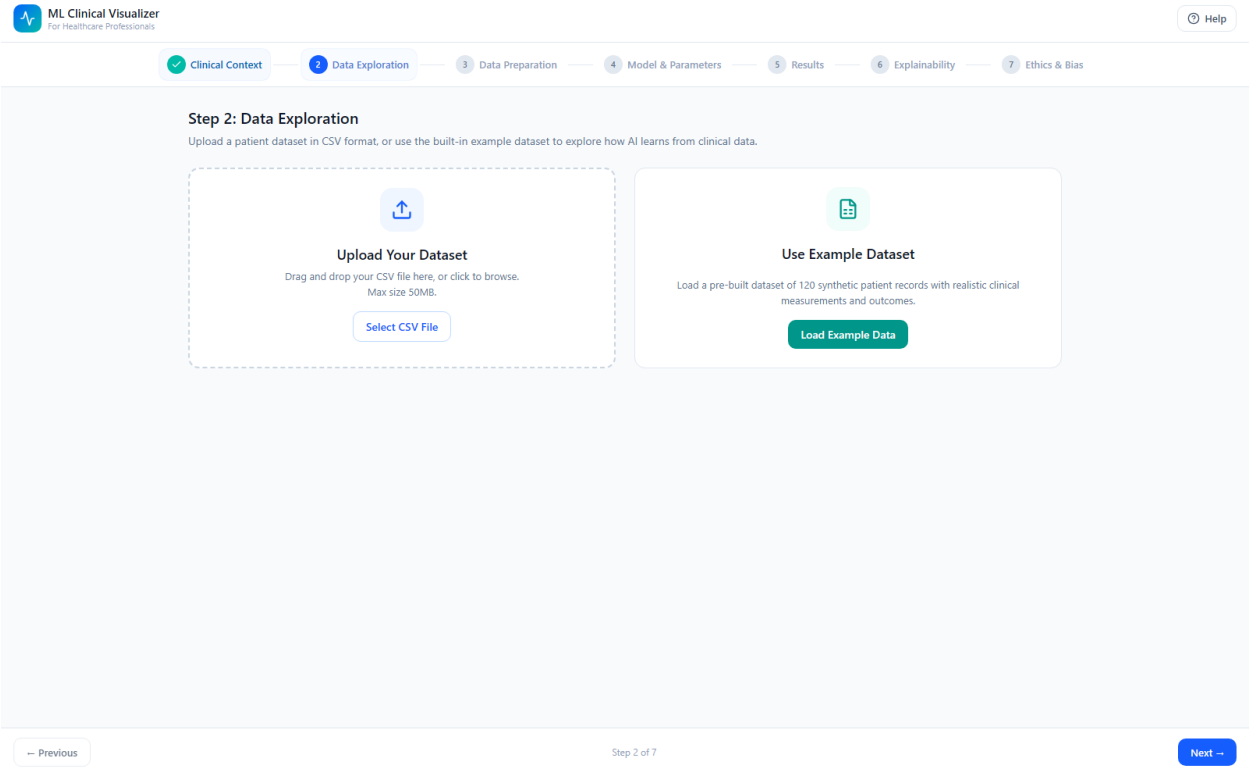


Figure: 01_step1_clinical_context_selected.png

Step 2 — Data upload and mapping

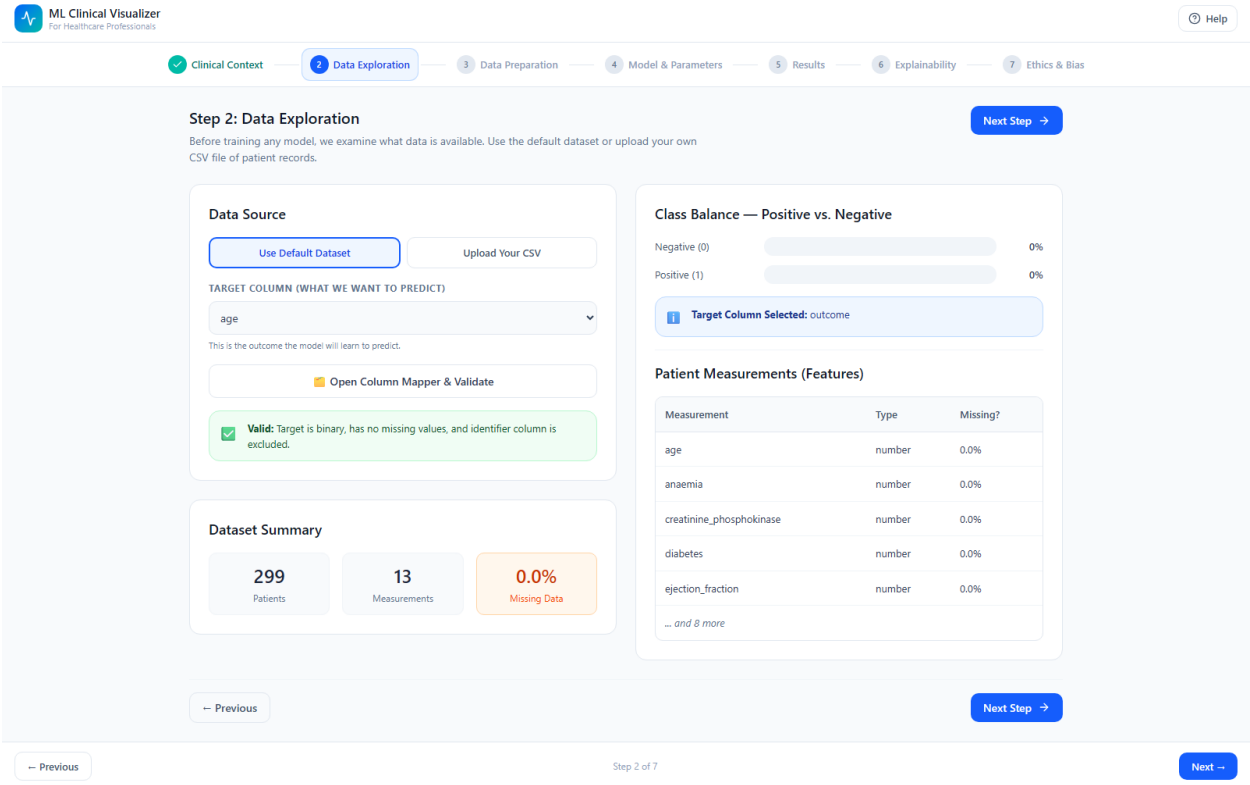


Figure: 02_step2_data_exploration_mapped.png

Step 3 — Data preparation applied

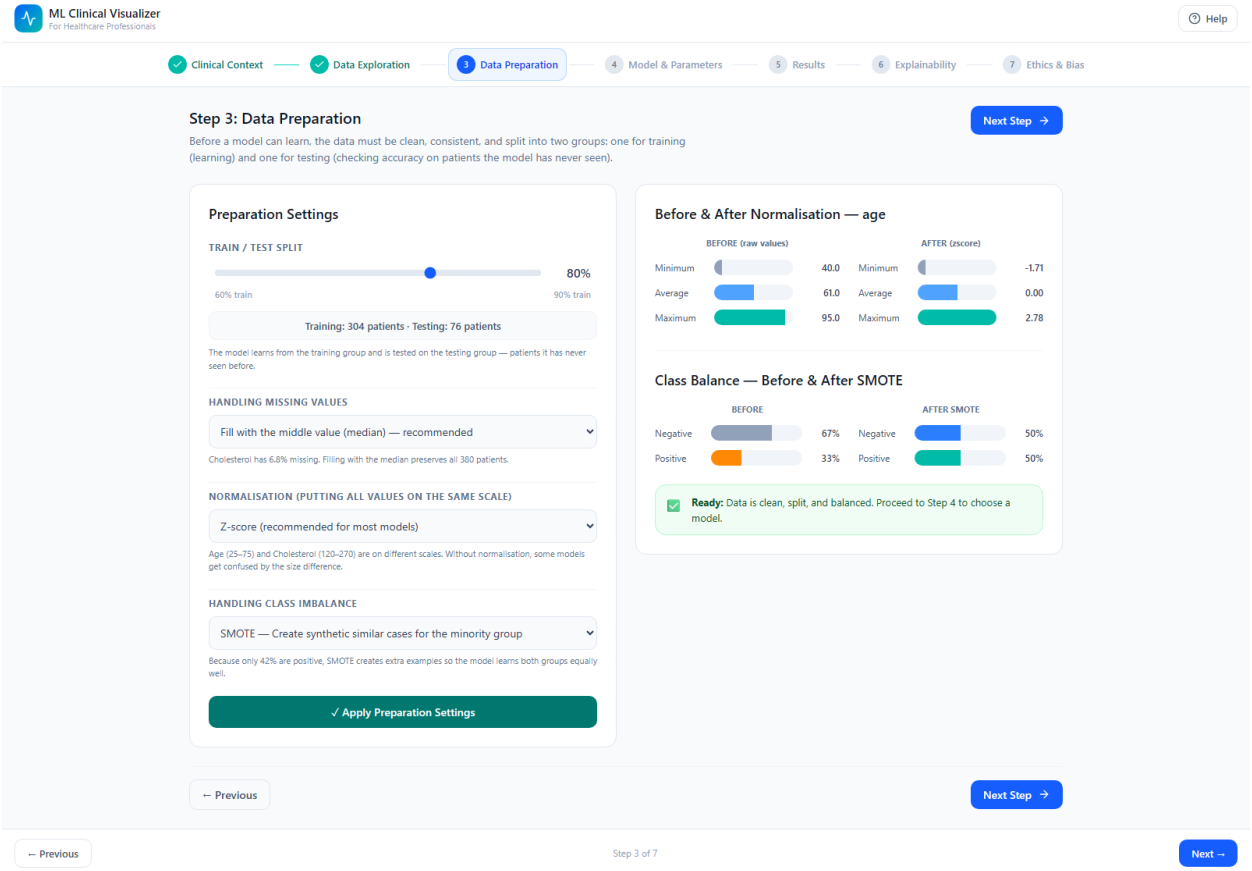


Figure: 03_step3_data_preparation_applied.png

Step 4 — Model and parameters

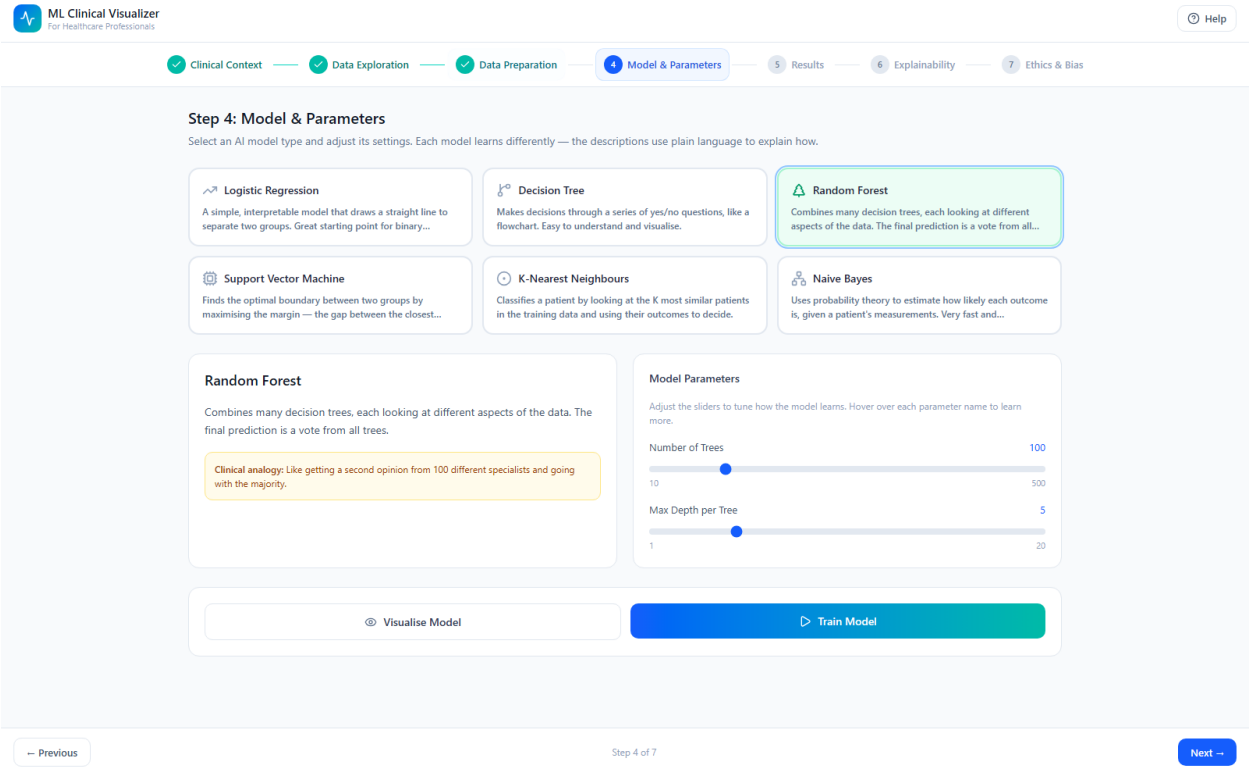


Figure: 04_step4_model_parameters.png

Step 5 — Results and metrics

ML Clinical Visualizer
For Healthcare Professionals

Help

Clinical Context — Data Exploration — Data Preparation — Model & Parameters — 5 Results — 6 Explainability — 7 Ethics & Bias

Step 5: Results — How Well Did the Model Perform?

Auto-retrain on change:

Choose a machine learning algorithm, adjust its settings, and train it on your patient data. Try different models and compare their accuracy side by side.

Choose Algorithm

KNN | SVM | Decision Tree | **Random Forest** | Logistic Reg. | Naive Bayes

Random Forest — A team of many decision trees. Each tree gets a random subset of data and votes. Highly accurate and robust to noise.

Parameters

NUMBER OF TREES

100

10 (fast) — 300 (stable)

More trees = more stable predictions, but slower to train. Each tree votes — the majority vote wins.

Train Model | + Compare

Auto-retrain queued (300ms debounce)...

Performance Metrics

ACCURACY	SENSITIVITY	SPECIFICITY
88%	85%	90%

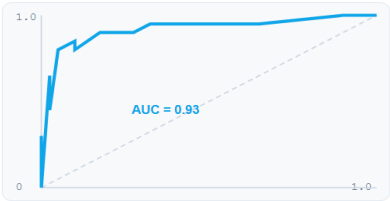
PRECISION	F1 SCORE	AUC
81%	83%	0.93

Green: meets target · Amber: caution zone · Red: below acceptable threshold

Metric	Green Threshold	Amber Threshold
ACCURACY	80%	70%
SENSITIVITY	75%	50%
SPECIFICITY	80%	70%
PRECISION	75%	60%
F1	75%	60%
AUC	85%	70%

Training latency: **192 ms** (target < 3000 ms)

ROC Curve — How Well Does the Model Separate the Two Groups?



A perfect model would go straight to the top-left corner. The blue curve comes from live backend training output. Current AUC: 0.93.

Confusion Matrix (TN / FP / FN / TP)

TN: **36**	FP: **4**
FN: **3**	TP: **17**

⚠ FN Warning: 3 positive patient(s) were missed. In screening use-cases this is clinically risky.

ℹ FP Info: 4 false alarm(s) may increase follow-up workload, but are often safer than missed positives.

Model Comparison

Model	Accuracy	Sensitivity ★	Specificity	AUC
Random Forest	88%	85%	90%	0.93

★ **Sensitivity** = how many of the truly positive patients did the model catch? This is the most important metric in screening.

← Previous | Next Step →

← Previous | Step 5 of 7 | Next →

Figure: 05_step5_results_metrics.png

Step 6 — Explainability

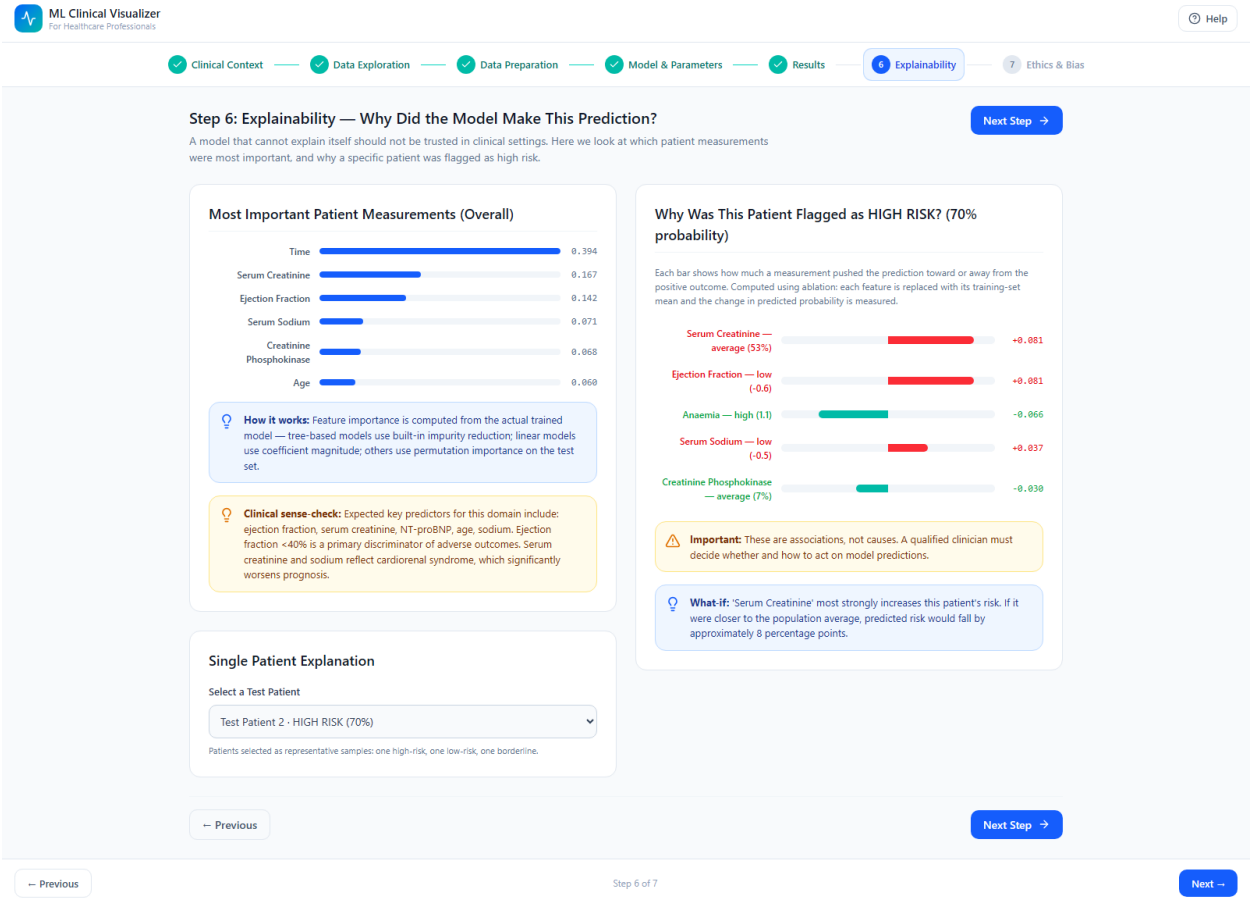


Figure: 06_step6_explainability.png

Step 7 — Ethics and bias

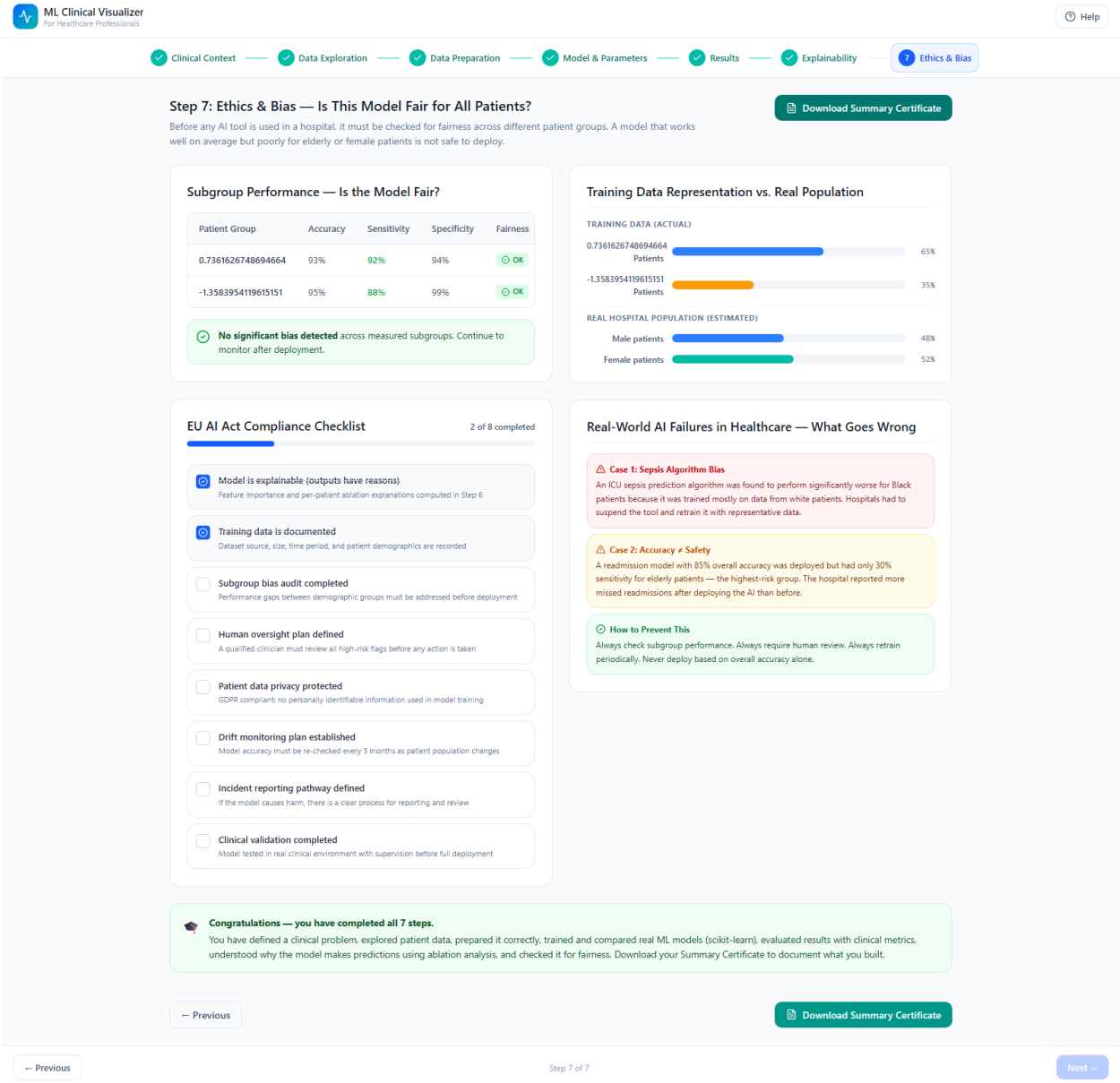


Figure: 07_step7_ethics_bias.png

7) Metrics

Metric	What to Measure	Target / Threshold	Current Result
Bias Detection Accuracy	Subgroup sensitivity gap rule	Hidden <=10 pp, appears >10 pp	PASS
Checklist Toggle	8 checklist controls + progress	All toggles work, progress visible	PASS
Certificate Content	Certificate endpoint result	Valid response for tested domains	PASS
Certificate Generation Time	API response timing	<10 seconds	PASS
End-to-End Flow	Steps 1–7 with fresh CSV	No crash, gates respected	PASS
Clinical Language Audit	Label readability	No raw snake_case exposed	PASS
Domain Coverage (Step 6–7)	Explainability/Bias multi-domain support	20 domains supported	PASS

8) Next Week Plan

1. Finalize submission package (report PDF/Word, evidence screenshots, QA note, Jira snapshots).
2. Push final Wiki updates and attach links to Sprint 4 deliverables.
3. Run final demo rehearsal script (5-minute showcase path).
4. Optional: align Playwright scripts with current UI labels to avoid false fails.

9) Retrospective Note (Keep / Improve / Try)

- Keep: Live branch verification before sprint sign-off reduced ambiguity and rework.
- Improve: Update automation tests immediately after UI/API refactors.
- Try: Enforce a fixed release checklist gate before reporting sprint completion.

Appendix — Evidence and Submission Notes

- This report follows the SENG430 weekly progress required sections format (1–9).
- All screenshots in this report were generated from the currently running system.
- Data used for validation: heart_failure_clinical_records_dataset.csv